Automobile Price Prediction Report

Data cleaning, exploratory analysis, and multiple linear regression modeling of the UCI Imports-85 dataset.

1. Executive Summary & Dataset Overview

This report covers the data cleaning, preprocessing, exploratory data analysis, and multiple linear regression modeling of the automobile price prediction dataset sourced from the UCI Machine Learning Repository (Imports-85). The original dataset contained 205 observations and 26 variables. Following the inspection and removal of invalid entries (encoded as ?), 199 valid observations remained for modeling.

Dataset Characteristic	Applied Result
Observations in original file	205 vehicles
Variables in original file	26 columns
Primary Explanatory Predictors	`horsepower`, `width`, `height`
Target Variable	`price` (numeric, USD)
Final Cleaned Observations	199 vehicles

2. Preprocessing & Missing-Value Assessment

Missing values and invalid placeholders (?) were addressed systematically. Rows containing missing data in the primary target or predictor subsets were dropped. Categorical attributes and price/horsepower metrics were converted to proper numeric formats to allow mathematical modeling without feature scaling distortion.

Observed Issue	Method Applied	Justification
Missing markers (?)	Replaced with NaN and dropped	Ensures data integrity for linear regression
Object data types	Converted via `to_numeric`	Required for continuous mathematical modeling
Feature Scaling	Not applied	Preserves direct unit interpretability of coefficients

3. Predictive Modeling Approach

Multiple Linear Regression was selected because the target variable is continuous and relationships between vehicle physical dimensions, power, and retail price are approximately linear. The dataset was split into an 80% training subset (159 observations) and a 20% testing subset (40 observations) using random seed 0.

- **Model Architecture:** Ordinary Least Squares Multiple Linear Regression
- **Train / Test Split:** 159 training / 40 testing samples
- **Evaluation Metric:** Coefficient of Determination (R^2 Score)

4. Applied Results & Model Evaluation

The primary model evaluating `horsepower`, `width`, and `height` against `price` yielded strong empirical results, explaining over 82% of price variance on unseen test data.

Model Configuration	Training R^2	Test R^2	Key Coefficients
Primary Model (Power & Dimensions)	0.748	**0.822**	Intercept: -109,617.10 Horsepower: +113.74 | Width: +1,490.94
Secondary Model (Weak Predictors)	0.004	**0.030**	Symboling, Compression-Ratio, Stroke (Poor linear correlation)

5. Conclusion & Limitations

The analysis demonstrates that vehicle power and physical dimensions are highly effective predictors of market price. The close alignment between training and test scores confirms robust generalization without overfitting. Future iterations could incorporate luxury branding metrics and non-linear feature interactions to further refine pricing accuracy.

1. Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.